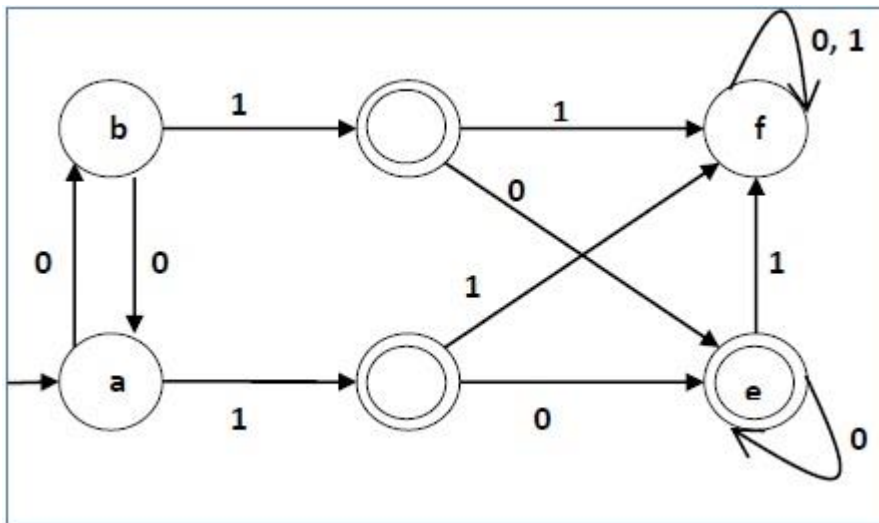


Example

Let us consider the following DFA –



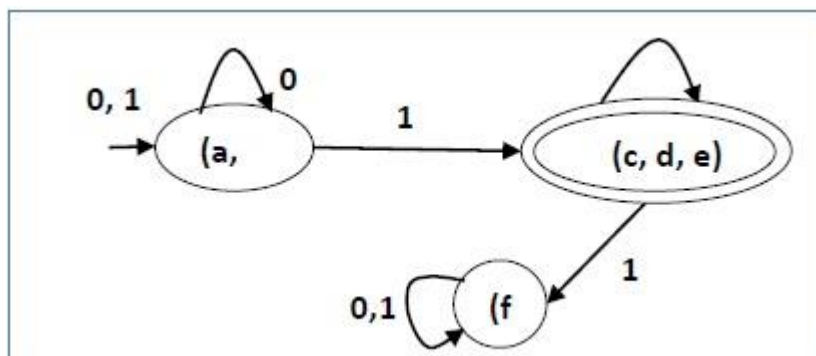
q	$\delta(q,0)$	$\delta(q,1)$
a	b	c
b	a	d
c	e	f
d	e	f
e	e	f
f	f	f

Let us apply the above algorithm to the above DFA –

- $P_0 = \{(c,d,e), (a,b,f)\}$
- $P_1 = \{(c,d,e), (a,b),(f)\}$
- $P_2 = \{(c,d,e), (a,b),(f)\}$

Hence, $P_1 = P_2$.

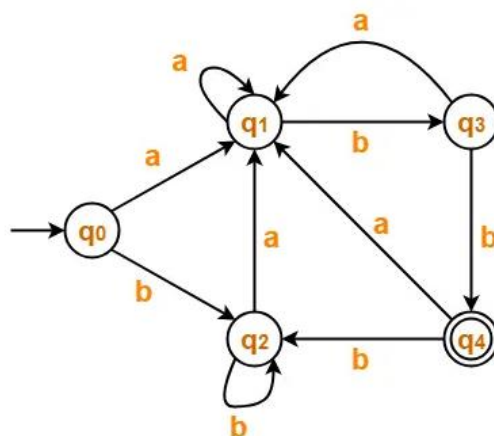
There are three states in the reduced DFA. The reduced DFA is as follows –



Q	$\delta(q,0)$	$\delta(q,1)$
(a, b)	(a, b)	(c,d,e)
(c,d,e)	(c,d,e)	(f)
(f)	(f)	(f)

Problem-01:

Minimize the given DFA-



Solution-

Step-01:

The given DFA contains no dead states and inaccessible states.

Step-02:

Draw a state transition table-

	a	b
→q0	q1	q2
q1	q1	q3
q2	q1	q2
q3	q1	*q4
*q4	q1	q2

Step-03:

Now using Equivalence Theorem, we have-

$$P_0 = \{ q_0, q_1, q_2, q_3 \} \{ q_4 \}$$

$$P_1 = \{ q_0, q_1, q_2 \} \{ q_3 \} \{ q_4 \}$$

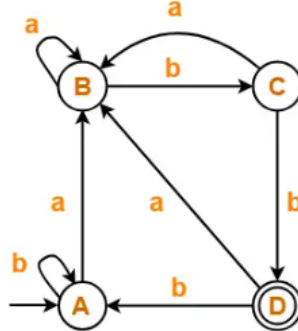
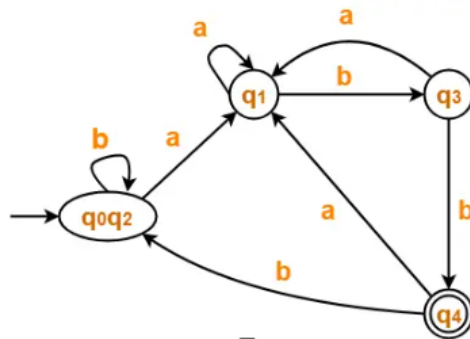
$$P_2 = \{ q_0, q_2 \} \{ q_1 \} \{ q_3 \} \{ q_4 \}$$

$$P_3 = \{ q_0, q_2 \} \{ q_1 \} \{ q_3 \} \{ q_4 \}$$

Since $P_3 = P_2$, so we stop.

From P_3 , we infer that states q_0 and q_2 are equivalent and can be merged together.

So, Our minimal DFA is-

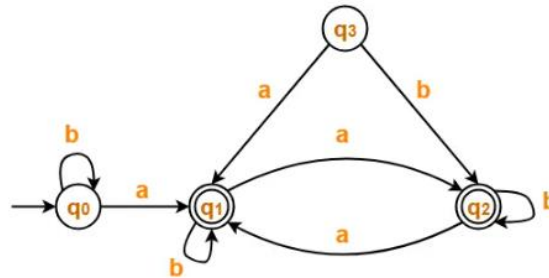


DFA Minimization | Problem-01 | Mir

Minimal DFA

Problem-02:

Minimize the given DFA-

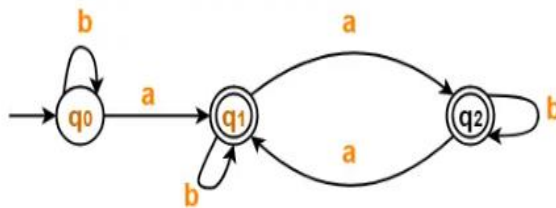


Solution-

Step-01:

- State q_3 is inaccessible from the initial state.
- So, we eliminate it and its associated edges from the DFA.

The resulting DFA is-



Step-02:

Draw a state transition table-

	a	b
$\rightarrow q_0$	*q1	q0
*q1	*q2	*q1
*q2	*q1	*q2

Step-03:

Now using Equivalence Theorem, we have-

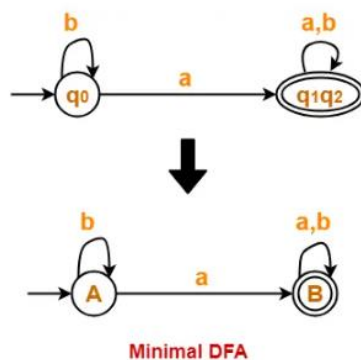
$$P_0 = \{ q_0 \} \{ q_1, q_2 \}$$

$$P_1 = \{ q_0 \} \{ q_1, q_2 \}$$

Since $P_1 = P_0$, so we stop.

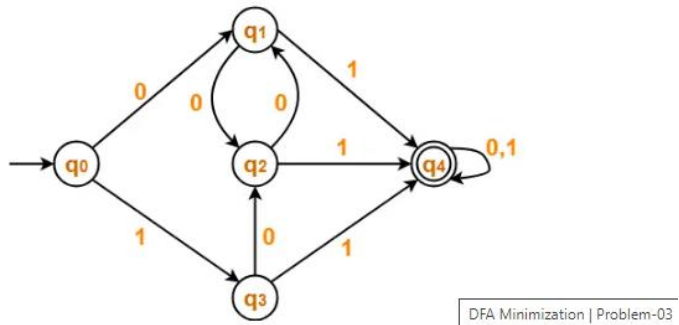
From P_1 , we infer that states q_1 and q_2 are equivalent and can be merged together.

So, Our minimal DFA is-



Problem-03:

Minimize the given DFA-



Solution-

Step-01:

The given DFA contains no dead states and inaccessible states.

Step-02:

Draw a state transition table-

	0	1
→q0	q1	q3

q1	q2	*q4
q2	q1	*q4
q3	q2	*q4
*q4	*q4	*q4

Step-03:

Now using Equivalence Theorem, we have-

$$P_0 = \{ q_0, q_1, q_2, q_3 \} \{ q_4 \}$$

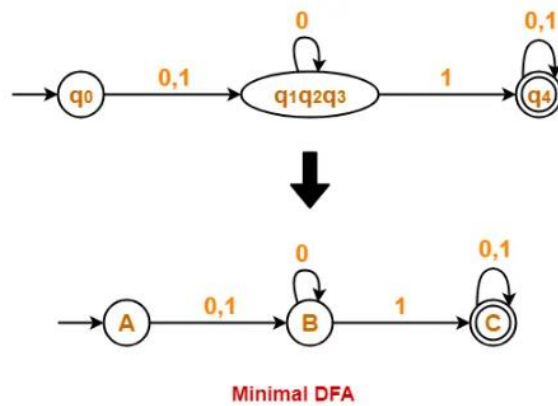
$$P_1 = \{ q_0 \} \{ q_1, q_2, q_3 \} \{ q_4 \}$$

$$P_2 = \{ q_0 \} \{ q_1, q_2, q_3 \} \{ q_4 \}$$

Since $P_2 = P_1$, so we stop.

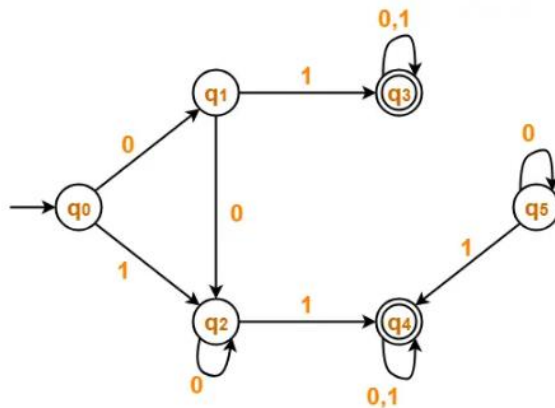
From P_2 , we infer that states q_1 , q_2 and q_3 are equivalent and can be merged together.

So, Our minimal DFA is-



Problem-04:

Minimize the given DFA-

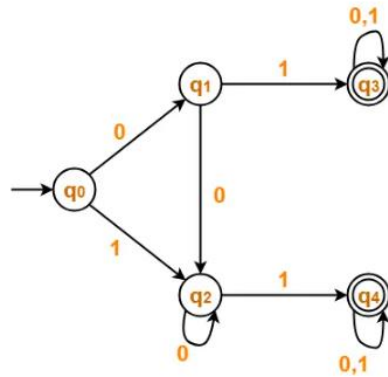


Solution-

Step-01:

- State q_5 is inaccessible from the initial state.
- So, we eliminate it and its associated edges from the DFA.

The resulting DFA is-



Step-02:

Draw a state transition table-

	0	1
→q0	q1	q2
q1	q2	*q3
q2	q2	*q4
*q3	*q3	*q3
*q4	*q4	*q4

Step-03:

Now using Equivalence Theorem, we have-

$$P_0 = \{ q_0, q_1, q_2 \} \{ q_3, q_4 \}$$

$$P_1 = \{ q_0 \} \{ q_1, q_2 \} \{ q_3, q_4 \}$$

$$P_2 = \{ q_0 \} \{ q_1, q_2 \} \{ q_3, q_4 \}$$

Since $P_2 = P_1$, so we stop.

From P_2 , we infer-

- States q_1 and q_2 are equivalent and can be merged together.
- States q_3 and q_4 are equivalent and can be merged together.

So, Our minimal DFA is-

